





Introduction of SONiX SN32F407 EVK Kit

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2026/7/30

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1. Recommended Products for Contest

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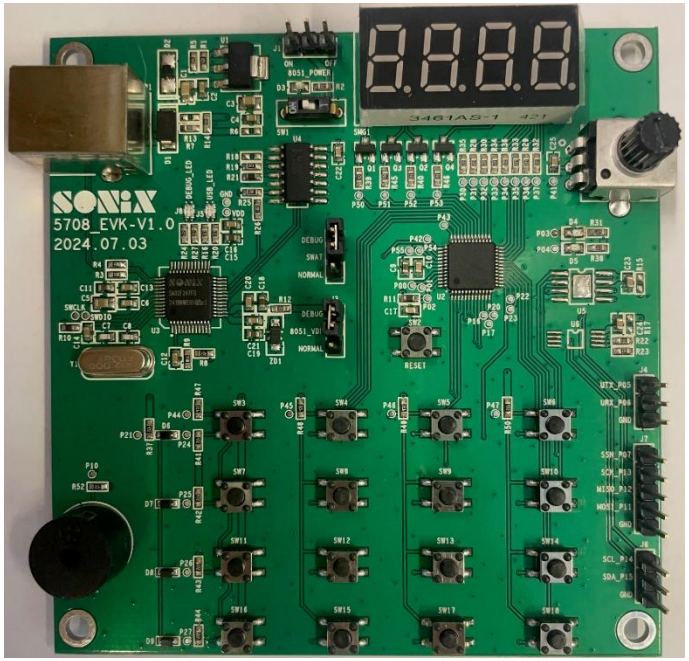
1. Recommended Products for Contest

For this contest, we would like to recommend several MCU series to develop, including:

- 8051 series: SN8F5708
- Cortex-M0 series: SN32F407
- Cortex-M4 series: SN34F788

1. Recommended Product for Contest

8051 series: SN8F5708



SN8F5708 EVK

Product Technical Specifications

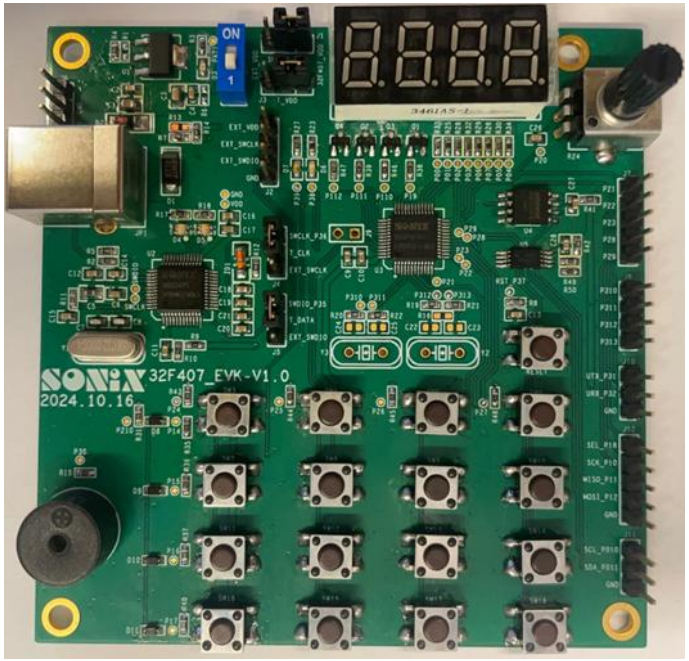
- ROM: 16 KB
- CPU Frequency: Up to 8 MHz flexible CPU clock
- Timer: 8/16-bit timer
- PWM: 16-bit PWM
- ADC: 12-bit SAR ADC
- Communication Interfaces: SPI, UART, I²C
- Operating Voltage: 1.8 V ~ 5.5 V

Datasheet: Click me

Demo Code: Click me

1. Recommended Product for Contest

Cortex-M0 series : SN32F407



SN32F407 EVK

Product Technical Specifications

- ROM: 32 KB
- CPU Frequency: Up to 60 MHz flexible CPU clock
- Timer: 16-bit timer
- PWM: 16-bit PWM
- ADC: 12-bit SAR ADC
- OPA: 1x~32x
- Communication Interfaces: SPI, UART, I^2C
- Operating Voltage: 1.8 V ~ 5.5 V

Datasheet: Click me

Demo Code: Click me

1. Recommended Product for Contest

Cortex-M4 series : SN34F788



SN34F788 STARTER KIT

Product Technical Specifications

- ROM: 512 KB
- CPU Frequency: Up to 192 MHz flexible CPU clock
- Timer: 16-bit timer
- PWM: 16-bit PWM
- ADC: 12-bit SAR ADC
- CAN 2.0
- LCD-TFT(8080)
- Communication Interfaces: SPI, UART, I^2C
- Operating Voltage: 2.4 V ~ 3.6 V

Datasheet: Click me

Demo Code: Click me

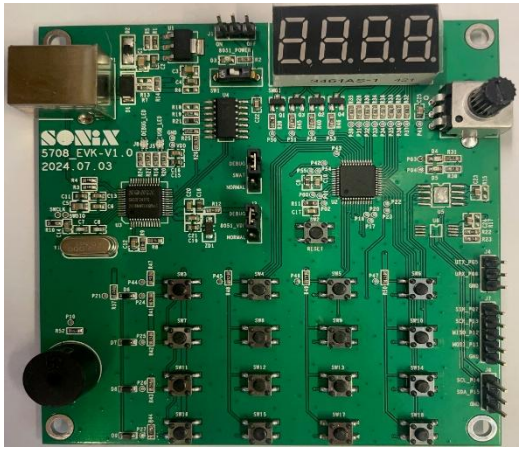
1. Recommended Product for Contest

Overview of MCU features

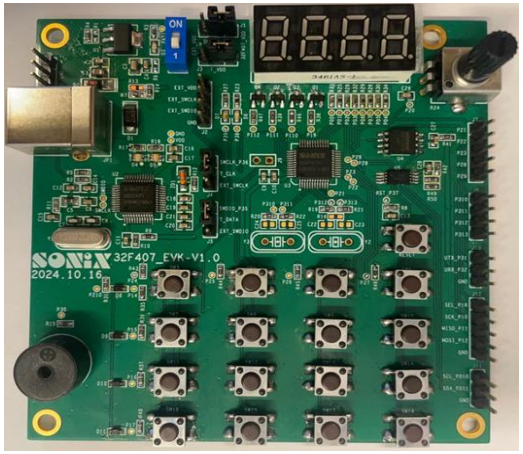
Overview of MCU features table

Chip	Operating Voltage	ROM (KB)	FCPU (MHz)	UART	SPI / I2S	I2C	CAN	Timer	PWM	ADC	CMP	OPA	LCD	ETHMAC	GPIO	PKG
SN8F5708	1.8V~5.5V	16	8	1	1 (SPI)	1	-	3	10	12	2	2	-	-	46	QFN48/LQFP48
SN32F407	1.8V~5.5V	32	60	2	1 (SPI)	1	-	3	12+4	16+6	4	3	-	-	46	QFN48/LQFP48
SN34F788	2.4V~3.6V	512	192	6	3	3	2	9	32+8	16	-	-	LCD-TFT (8080)	Y	55	LQFP64

Legend: - : none Y: yes



SN8F5708 EVK



SN32F407 EVK



SN34F788 STARTER KIT

2. Technical Support

We will provide technical support as follows:

1. Review the schematic of MCU if the teams design your own PCB.

2. Support the SDK of the recommended MCU series.

3. Provide direct technical support and real-time Q&A assistance in Zalo group.

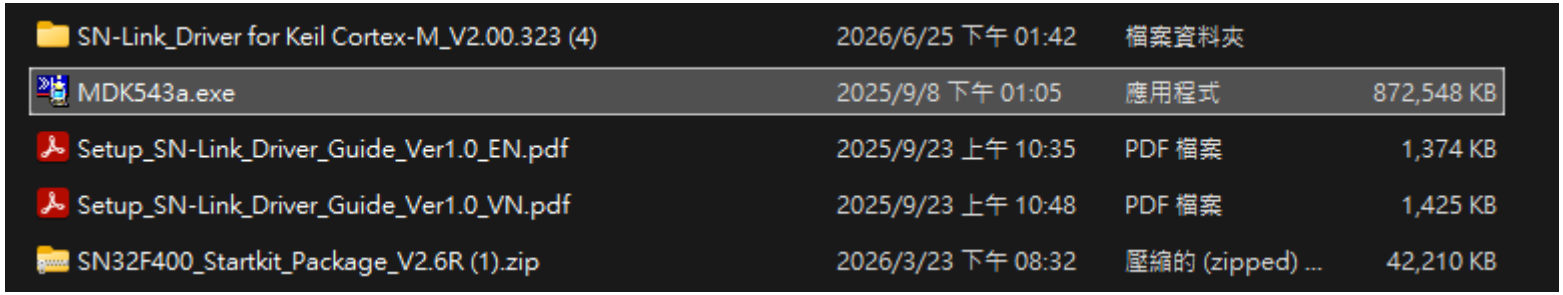
3. Software Guide

The software we need install:

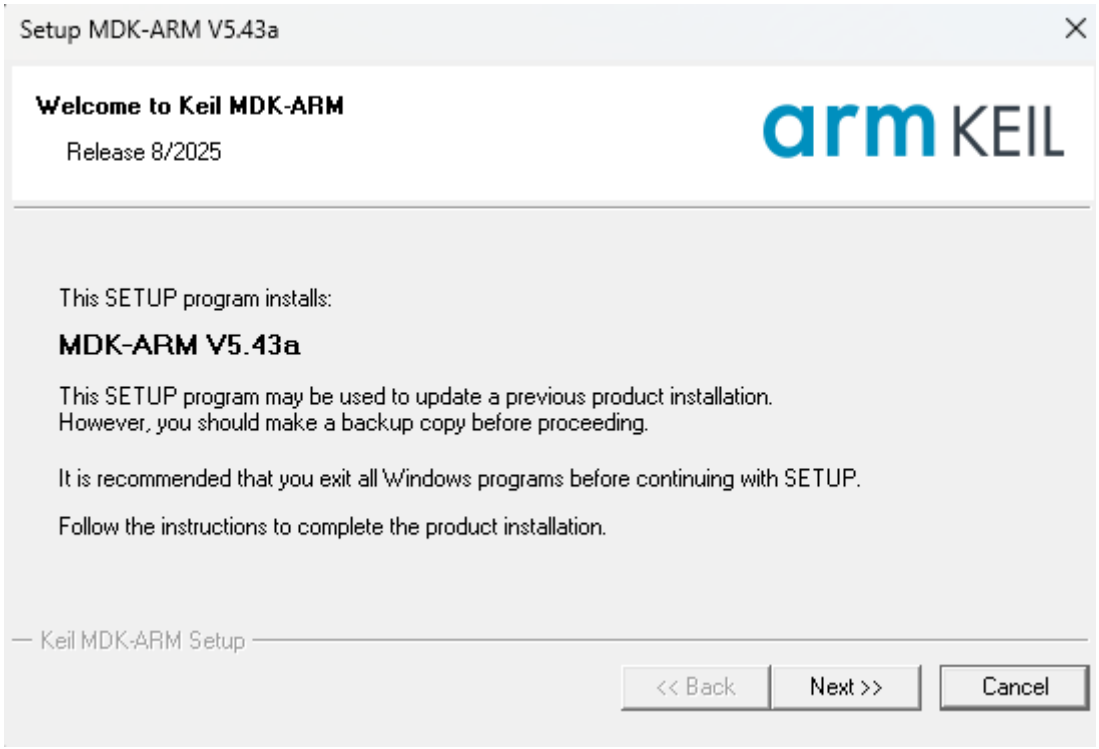
1. MDK-Arm Keil
2. SN-Link_Driver for Keil Cortex-M
3. SONiX_SN32F4_DFP pack

3. Software Guide
MDK-Arm Keil

1. Click file MDK543a.exe with administrator

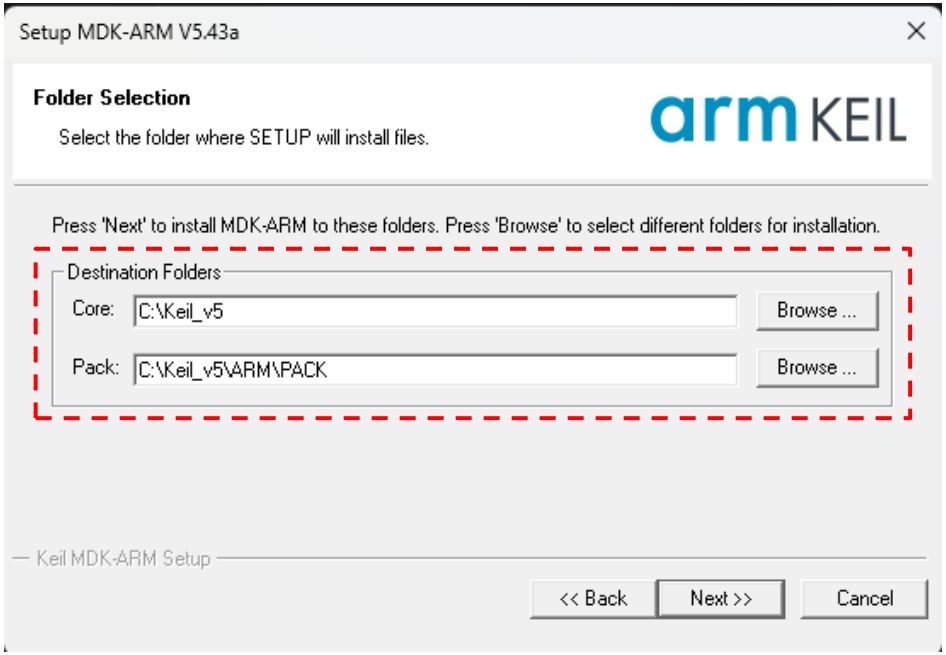


2. Click Next


Setup MDK-ARM V5.43a
Welcome to Keil MDK-ARM
Release 8/2025
arm KEIL
This SETUP program installs:
MDK-ARM V5.43a
This SETUP program may be used to update a previous product installation.
However, you should make a backup copy before proceeding.
It is recommended that you exit all Windows programs before continuing with SETUP.
Follow the instructions to complete the product installation.
Keil MDK-ARM Setup
<< Back
Next >>
Cancel

3. Software Guide
MDK-Arm Keil

3. Click Next, the system will prompt you to select an installation path and Pack library Path



Note: **We recommend using the system default installation path to ensure smooth driver installation. If you choose a custom path, you must select the same path again when installing the driver.**

3. Software Guide
MDK-Arm Keil

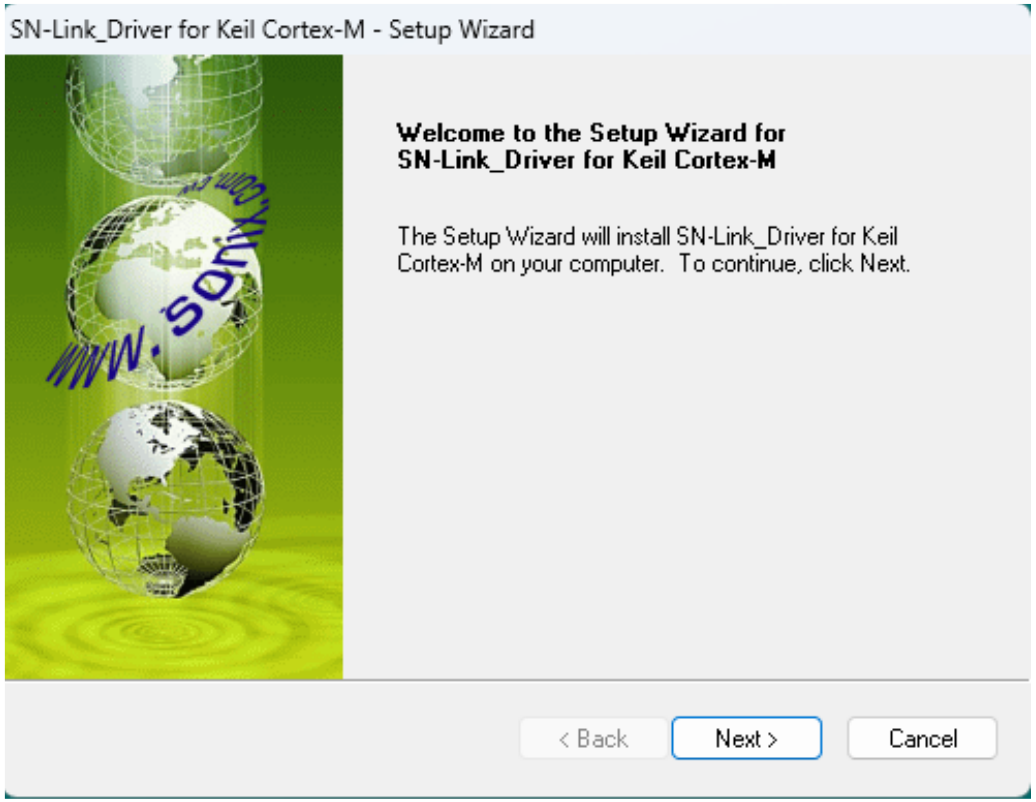
4. Continue clicking Next until the installation of Keil MDK-Arm starts



➡ Install Keil MDK-Arm guide has been completed!

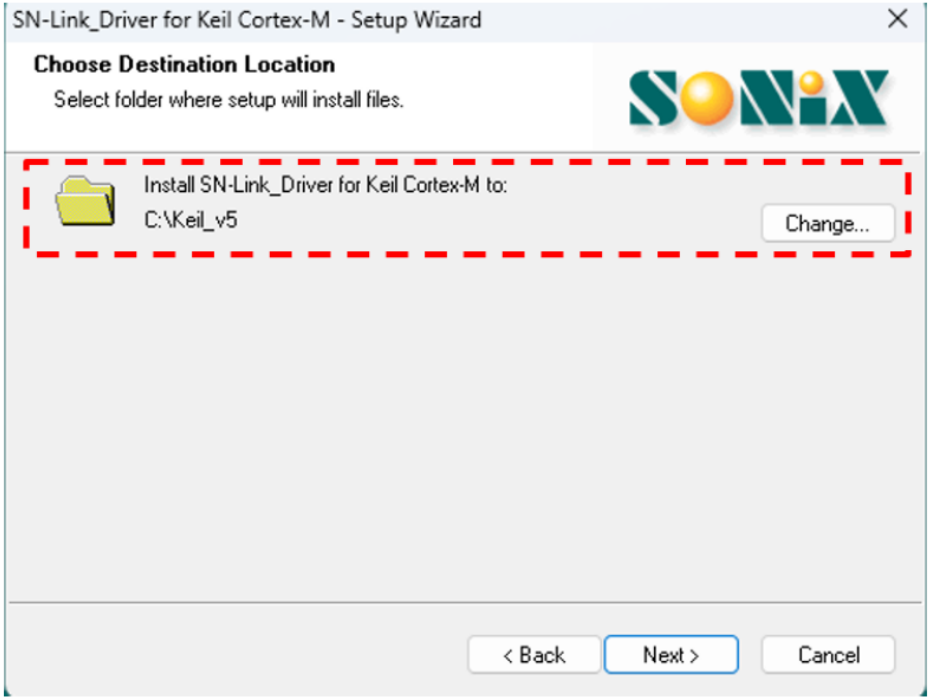
3. Software Guide
SN-Link_Driver for Keil Cortex-M

1. Start install SN-Link Driver for Keil Cortex-M.



3. Software Guide
SN-Link_Driver for Keil Cortex-M

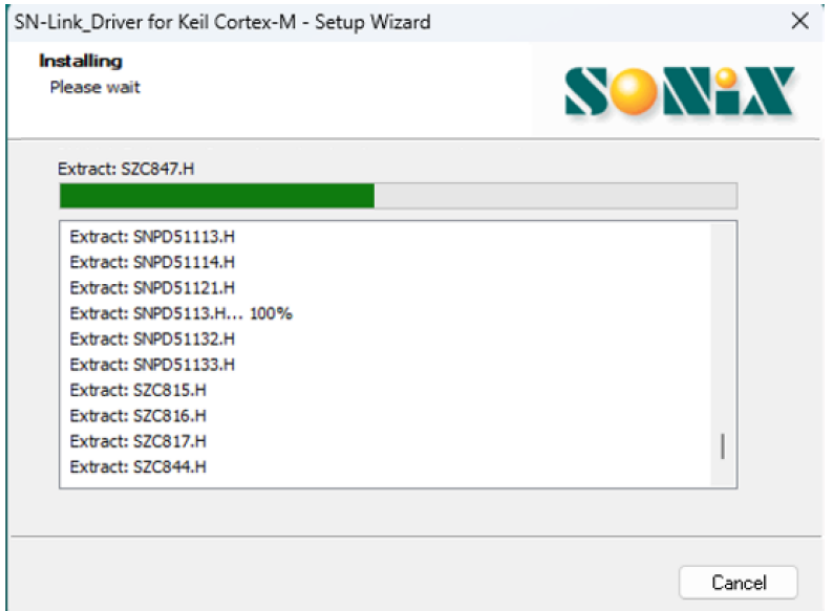
2. Click Next, the system will prompt you to select an installation path.



Note: We recommend using the system default installation path for the driver. If a custom path is selected, it must be the same as the Keil IDE installation path.

3. Software Guide
SN-Link_Driver for Keil Cortex-M

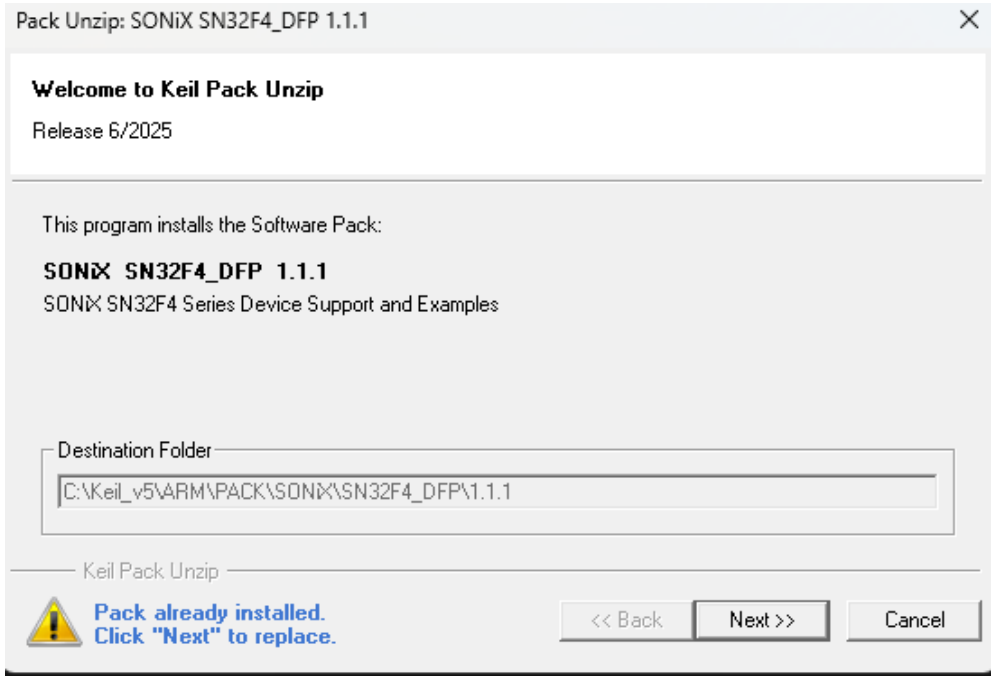
3. Continue clicking Next until the installation of SN-LINK Driver starts



➡ Install SN-Link_Driver for Keil Cortex-M guide has been completed!

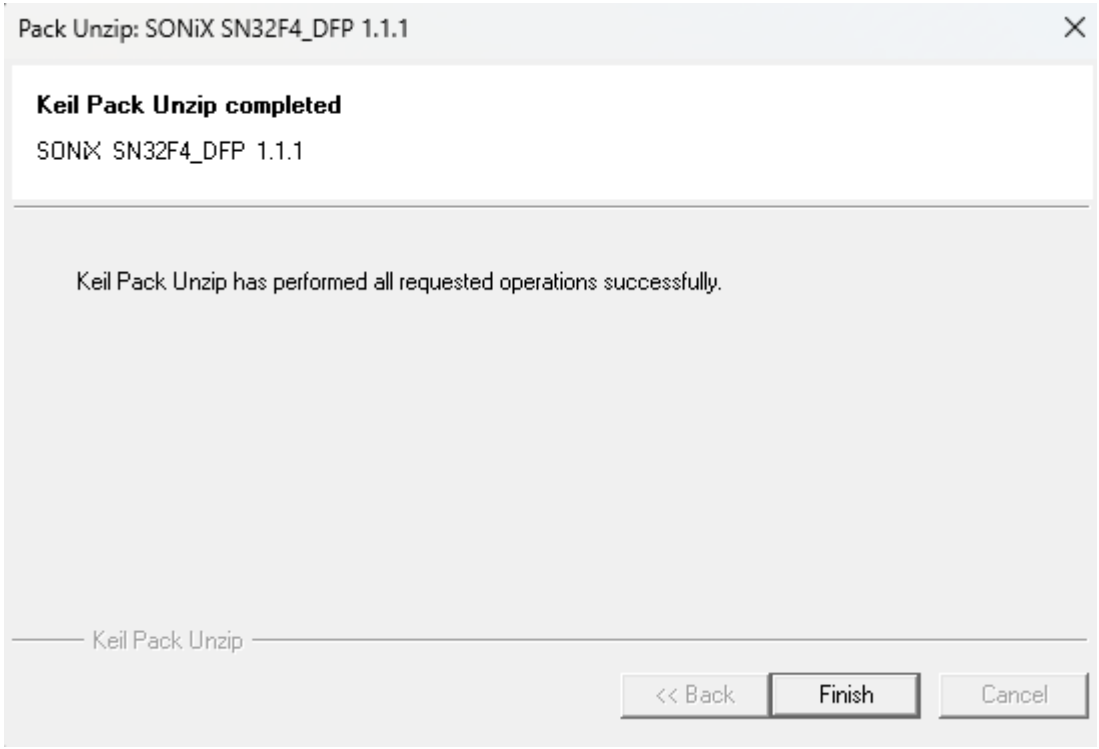
3. Software Guide
SONiX_SN32F4_DFP pack

1.Click next to install SONiX_SN32F4_DFP pack until download done.



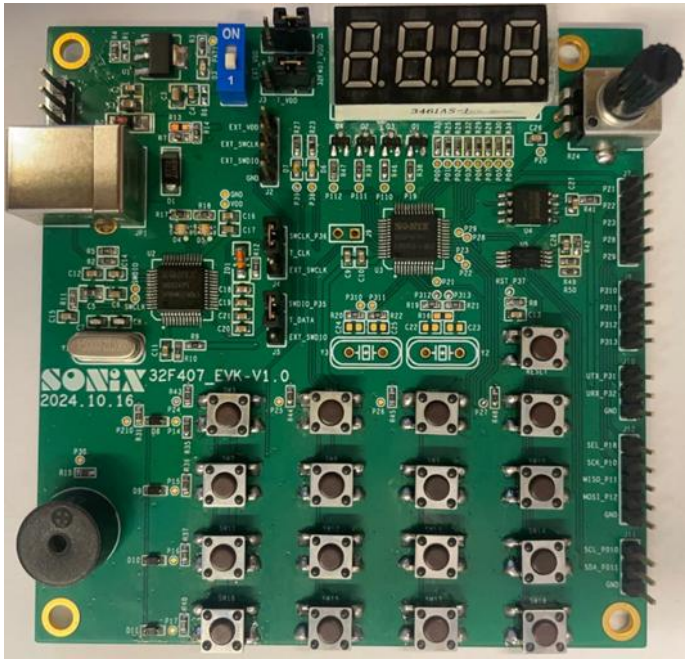
3. Software Guide
SONiX_SN32F4_DFP pack

2. Install SONiX_SN32F4_DFP pack has been completed!



4. Hardware Guide

SN32F407 EVK



SN32F407 EVK

Product Technical Specifications

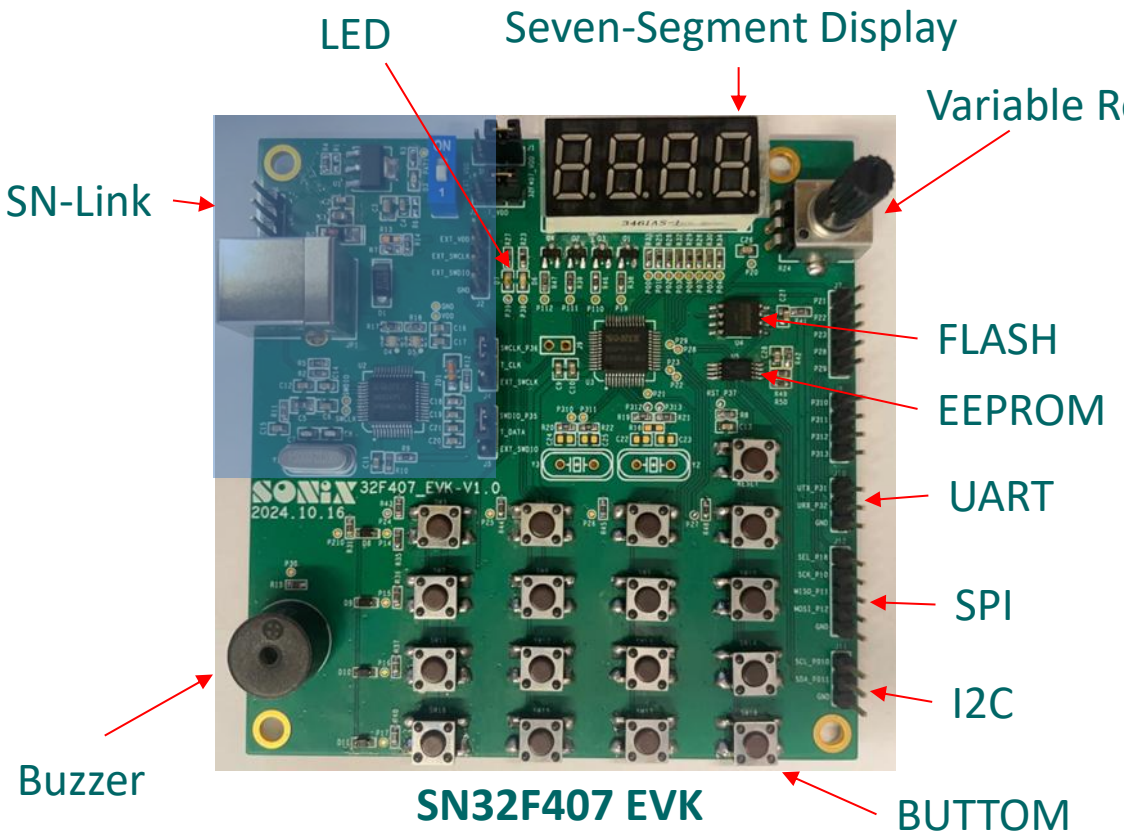
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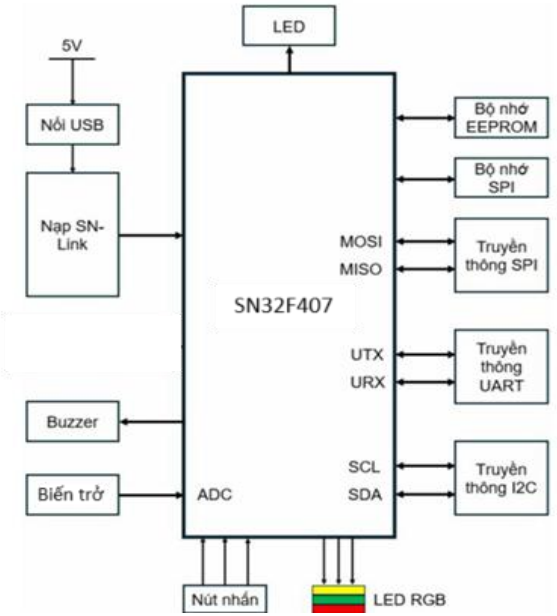
Datasheet: Click me

Demo Code: Click me

4. Hardware Guide

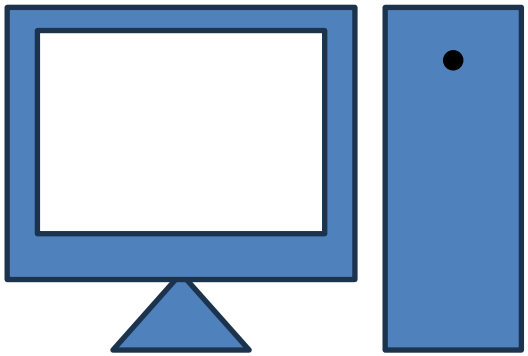
SN32F407 EVK





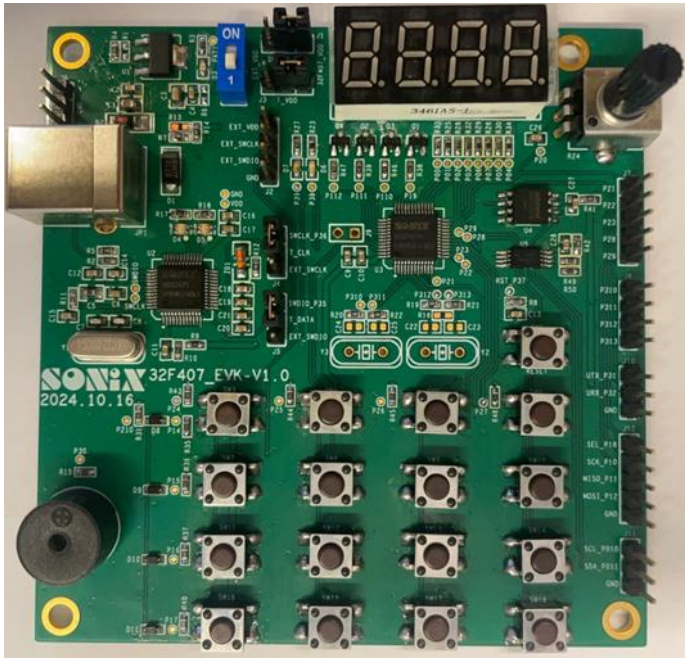
4. Hardware Guide

SN32F407 EVK



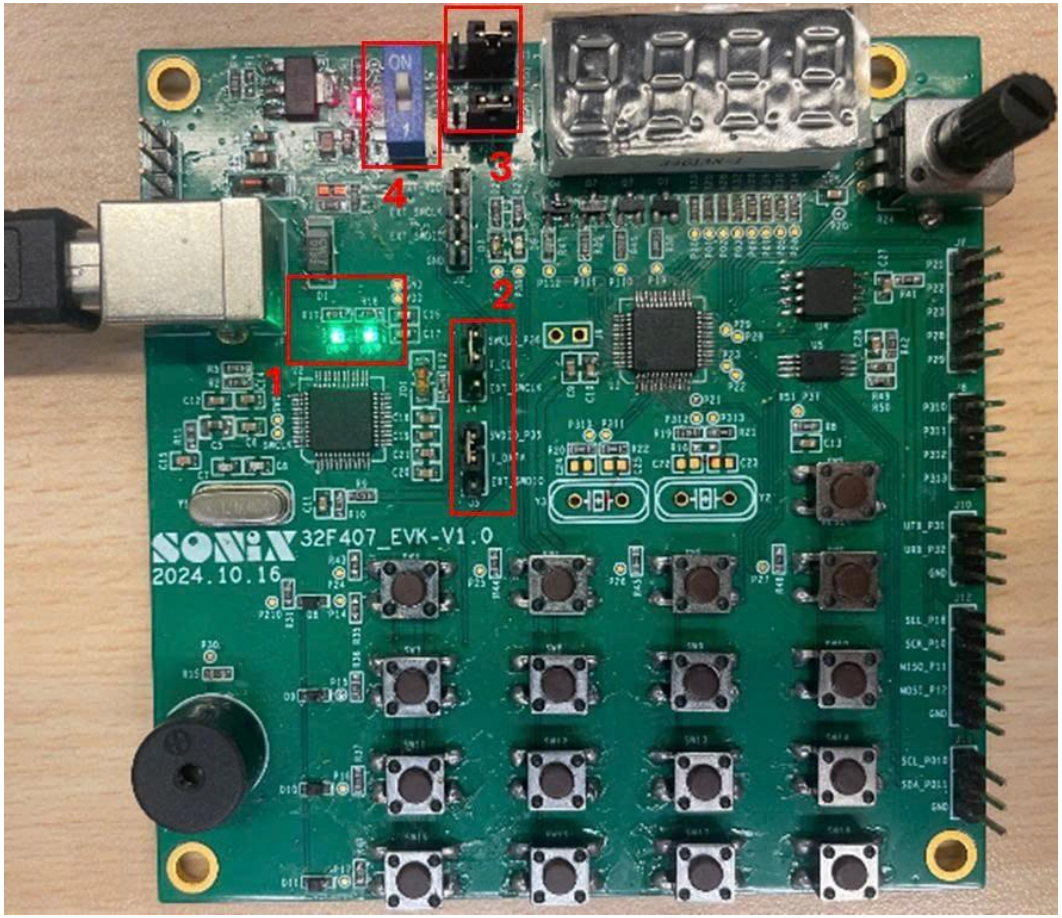
PC

USB Cable to PC



4. Hardware Guide

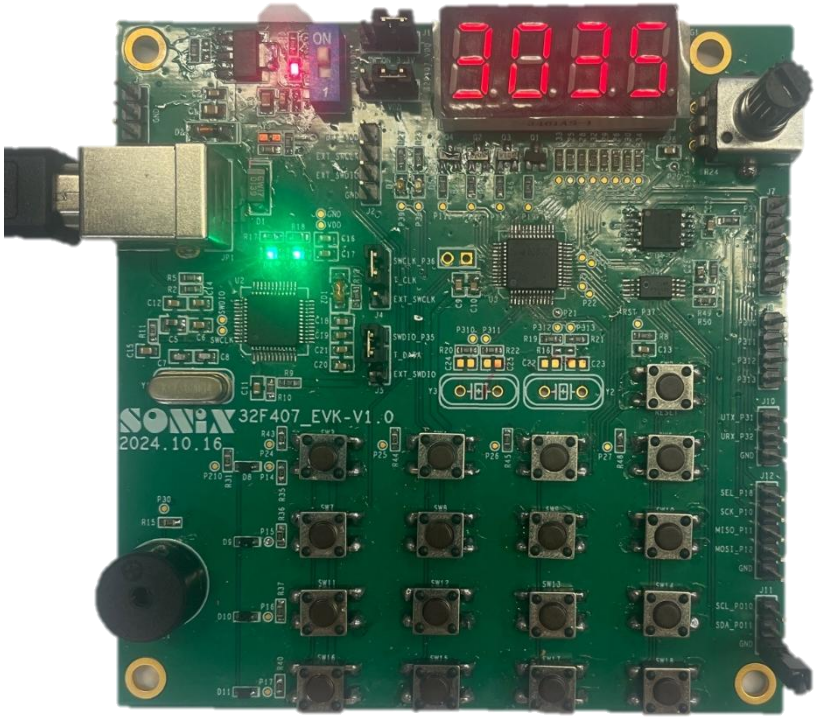
SN32F407 EVK



5. Demo Code Training

DEMO

1. Control LED BLINK
2. Control LED 7 SEG
3. CHECK VOLTAGE WITH ADC



5. Demo Code Training

DEMO

1. Control LED BLINK

```
77      while (1)
78      {
79          __WDT_FEED_VALUE;                   // feed dog
80
81          SET_LED0_ON;                            //set led on
82          SET_LED1_ON;
83
84          UT_DelayNms(100);                   //delay about 100ms
85
86          SET_LED0_OFF;                           //set led off
87          SET_LED1_OFF;
88
89          UT_DelayNms(100);                   //delay about 100ms
90      }
```

5. Demo Code Training

DEMO

2. Control LED 7 SEG

```
79      while (1)
80      {
81          __WDT_FEED_VALUE;
82
83          read_key = single_level_detect();                    //read key action
84          if(read_key)          //key push event
85          {
86              increase_value = increase_value < 9999 ? increase_value+1 : 0;
87              Digital_DisplayDEC(increase_value);
88          }
89          Digital_Scan();
90          UT_DelayNx10us(100);
91      }
92  }
```

5. Demo Code Training

DEMO

3. CHECK VOLTAGE WITH ADC

```
84      while (1)
85      {
86          __WDT_FEED_VALUE;
87          if(timer_1ms_flag)
88          {
89              timer_1ms_flag = 0;
90              Digital_Scan();
91
92
93              SN_ADC->ADM_b.ADS = 1;       //adc sample 1tims/ms
94
95          }
96          if(SN_ADC->ADM_b.EOC)    //end of convert
97          {
98              adb = SN_ADC->ADB & 0xfff;
99              SN_ADC->ADM_b.ADS = 1;       //START ADC SAMPLE
100
101             adc_sample += (int16_t)(adb - adc_sample) >> 3;     //low pass filter
102
103             Digital_DisplayDEC(adc_sample);
104         }
105
106     }
```

The End